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EUROPEAN UNION AVIATION SAFETY AGENCY
DIRECTORATE OF AERODROMES & AIR TRAFFIC MANAGEMENT

OPERATIONAL EXEMPTION & SAFETY CASE
EASA/ADR–EXEMP/2024–009
Issued: 04 November 2024

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Subject: Operational Exemption & Safety Assessment for Urgent Runway Rehabilitation with Displaced Threshold and Reduced Runway End Safety Area (RESA)

REF. PUBLICATIONS & STATUTORY JURISDICTION:

- **Regulation (EU) 2018/1139 (Basic Regulation):** Article 71(1) (Flexibility Provisions & Urgent Operational Exemptions).
- **Regulation (EU) No 139/2014 (Aerodromes):** Annex III (Part-ADR.OR) Point ADR.OR.B.040 (Changes requiring prior approval) and Annex IV (Part-ADR.OPS) Point ADR.OPS.B.070 (Aerodrome Works Safety).
- **CS-ADR-DSN (Certification Specifications):** CS-ADR-DSN.B.035 (Runway End Safety Areas - RESA) & CS-ADR-DSN.B.075 (Threshold Displacement & Runway Marking Standards).
- **ICAO Doc 9981 (PANS-Aerodromes):** Part II, Chapter 3 (Managing Aerodrome Operations during Non-Standard Works).

APPLICABILITY:

- National Competent Authorities (NCAs / Direction de la Sécurité de l'Aviation Civile - DSAC).
- Certified Aerodrome Operators under Regulation (EU) No 139/2014 managing active runway maintenance.
- Air Navigation Service Providers (ANSPs / Control Towers) and Commercial Airline Flight Operations (Flight Dispatch).

1. EXECUTIVE SUMMARY & OPERATIONAL PURPOSE

This document constitutes the official EASA Operational Exemption and Safety Assessment Synthesis issued pursuant to **Article 71(1) of Regulation (EU) 2018/1139**. It establishes the mandatory regulatory conditions, recalculated declared distances, and compensatory safety controls permitting continued commercial flight operations during urgent structural rehabilitation of subterranean drainage infrastructure beneath the Runway 26 physical strip.

Where full compliance with CS-ADR-DSN physical runway characteristics cannot be maintained during active civil engineering works, European aerodrome operators must formally demonstrate an **Equivalent Level of Safety (ELOS)** through comprehensive risk mitigation, stringent meteorological operating restrictions, and formal inter-service coordination.

2. STATUTORY BASIS (ARTICLE 71 FLEXIBILITY PROVISIONS)

Pursuant to Article 71(1) of Regulation (EU) 2018/1139, Member States and aerodrome operators may grant temporary exemptions from applicable European technical requirements in the event of *unforeseen urgent operational circumstances* or *urgent operational needs*, provided that:

- An Equivalent Level of Safety (ELOS) is achieved through validated compensatory measures;
- The temporary exemption does not exceed a statutory duration of **two months (60 calendar days)** without formal Commission review;
- Adverse commercial discrimination between aircraft operators is strictly prevented; and
- A formal Aeronautical Information Publication (AIP) Supplement and trigger NOTAM are disseminated internationally.

3. DESCRIPTION OF URGENT INFRASTRUCTURE DEFECT

Routine airside ground-penetrating radar (GPR) surveillance and visual pavement inspections revealed subterranean structural subsidence affecting a major reinforced concrete stormwater culvert crossing beneath Runway 08/26 at chainage 2,240m from the physical threshold of Runway 08. The defect compromises subgrade structural load-bearing capacity (PCN) in the western overrun zone. Immediate excavation and deep structural reinforcement are required to prevent pavement collapse under widebody aircraft dynamic impact loads.

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4. PHYSICAL GEOMETRY & TEMPORARY RUNWAY RECONFIGURATIONS

To isolate the active excavation zone from landing and departing aircraft, the aerodrome operator must implement an uncompromised physical safeguarding buffer in accordance with Point ADR.OPS.B.070:

- **Displaced Landing Threshold (Runway 26):** Threshold 26 is temporarily displaced forward into the runway by **320 meters**. Arriving aircraft touch down beyond the works area.
- **Temporary Runway End Safety Area (RESA):** In lieu of the standard 240m recommended RESA, a temporary certified RESA of **120 meters length by 90 meters width** is established beyond the temporary displaced threshold, compliant with the ICAO Code 4 minimum standard (90m).
- **Excavation Buffer:** The active work site is enclosed by a frangible, continuous barrier line located 60 meters west of the temporary threshold, maintaining complete isolation from the active pavement.

Recalculated Declared Distances (Comparison Baseline)

The temporary displacement reduces the Landing Distance Available (LDA) on Runway 26 while preserving full Take-Off Run Available (TORA) on Runway 08:

Runway Designator	TORA (Take-Off Run)	TODA (Take-Off Dist.)	ASDA (Accel.-Stop)	LDA (Landing Dist.)	Operational Remarks
RWY 26 (Standard)	3,000 m	3,150 m	3,000 m	3,000 m	Normal unrestricted operations.
RWY 26 (Temporary)	**2,680 m**	**2,830 m**	**2,680 m**	**2,680 m**	**Displaced by 320 m. Reduced LDA.**
RWY 08 (Standard)	3,000 m	3,150 m	3,000 m	3,000 m	Normal unrestricted operations.
RWY 08 (Temporary)	**3,000 m**	**3,000 m**	**2,680 m**	**2,680 m**	**Overrun protection / Reduced ASDA.**

5. HAZARD IDENTIFICATION & RISK ASSESSMENT (HIRA)

In accordance with ICAO Doc 9981 safety methodology, the operator's safety management system (SMS) assessed all operational hazards arising from the reduced runway dimensions:

Identified Hazard	Potential Operational Consequence	Initial Risk	Mandated Mitigation	Residual Risk
Landing Overrun (RWY 26)	Aircraft exits reduced LDA into excavation zone.	**High (4A)**	Dry runway limitation; max 15 kt crosswind cap; 120m certified RESA.	**Low (1B)**
Landing Undershoot (RWY 26)	Aircraft touches down short of displaced threshold.	**High (4B)**	PAPI realigned to 3.0°; prominent visual chevrons and green threshold lights.	**Low (1A)**
Foreign Object Debris (FOD)	Construction debris ingested by jet turbine engines.	**Medium (3B)**	Broom-clean standard; suction sweeper patrols every 3 hours; sealed barriers.	**Low (1A)**
Visual Confusion at Night	Flight crew mistakes decommissioned pavement for runway.	**Medium (3C)**	Unserviceability crosses (X); yellow flashing beacons; extinguished centerline.	**Low (1A)**

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6. MANDATORY COMPENSATORY SAFETY MEASURES (ELOS ENFORCEMENT)

To satisfy the statutory requirement for an Equivalent Level of Safety under Article 71(1), the following five operational compensatory measures are legally binding upon all users. Failure to comply with any individual measure shall trigger immediate suspension of modified runway operations:

CONDITION 1: METEOROLOGICAL OPERATING LIMITATIONS (DRY RUNWAY ONLY)

Provided that the runway surface condition is certified as **DRY** (nil standing water, nil slush, nil ice), commercial turbojet landings on Runway 26 are permitted. Whenever precipitation induces a contaminated or wet surface condition (braking action below 'Good'), commercial landings on Runway 26 shall be suspended immediately.

CONDITION 2: MAXIMUM CROSSWIND LIMITATION (15 KNOTS CAP)

The maximum allowable crosswind component for landing on Runway 26 during the exemption period is capped at **15 knots (including gusts)**, regardless of higher aircraft manufacturer crosswind certification limits, to minimize lateral landing rollout deviation.

CONDITION 3: ENHANCED RESCUE AND FIRE FIGHTING SERVICES (RFFS STANDBY)

During active construction hours, one heavy RFFS Category 7 Rapid Intervention Vehicle (RIV) shall be positioned at **Standby Status at Taxiway Bravo Holding Point**, maintaining an immediate response capability under 60 seconds to both runway ends.

CONDITION 4: VISUAL AIDS & PRECISION APPROACH PATH INDICATOR (PAPI) REALIGNMENT

In lieu of standard approach lighting to physical Threshold 26, the 4-unit Precision Approach Path Indicator (PAPI) has been flight-checked and recalibrated to an approach slope angle of **3.0 degrees**, referenced strictly to the temporary displaced threshold. The decommissioned 320m runway section is marked with prominent yellow chevrons and permanent unserviceability markings.

CONDITION 5: FRANGIBILITY, FOD CONTROL & CONTINUOUS SWEEPER PATROLS

The works perimeter must be demarcated by a continuous line of low-mass frangible barriers ballasted with dry sand, fitted with red/white aviation safety flags and omnidirectional flashing yellow beacons (minimum 40 candelas). Airside operations shall execute a motorized suction sweeper inspection every three hours, certifying the surface 100% broom-clean.

7. CONNECTING WORDS FOR AVIATION RULES (LINGUISTIC PROTOCOL)

In international safety cases and operational directives, precision connectors define legal boundaries. The following terms govern this exemption:

- **Provided that:** Establishes an inescapable prerequisite. *(e.g., 'Operations may proceed provided that the friction coefficient exceeds 0.50.')*
- **Subject to:** Subordinates an authorization to regulatory oversight. *(e.g., 'This approval is subject to daily DSAC compliance audits.')*
- **In lieu of:** Identifies an approved alternative substituting for standard compliance. *(e.g., 'A 120m RESA is established in lieu of the standard 240m buffer.')*
- **Notwithstanding:** Overrides general provisions under exceptional conditions. *(e.g., 'Notwithstanding published aerodrome manual hours, night culvert paving is authorized.')*

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EASA/ADR-EXEMP/2024-009 • RUNWAY 26 EXEMPTION & SAFETY CASE **SECTION 8 & 9: DISSEMINATION & AUDIT**

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8. AERONAUTICAL NOTIFICATION & AIRLINE OPERATIONAL BRIEFINGS

To ensure absolute situational awareness across international airline operators, flight dispatchers, and multinational flight crews, the aerodrome operator must implement the following multi-tiered notification workflow:

A. AIP Supplement (SUP) Publication

In accordance with ICAO Annex 15 standards, a temporary AIP Supplement (SUP 014/24) was published 28 days prior to the effective date, providing flight management computer (FMC) database coding amendments, revised declared distances, and updated aerodrome ground movement charts.

B. Trigger NOTAM Telecommunication

A Class 1 trigger NOTAM (Series A) was promulgated via the international Aeronautical Fixed Telecommunication Network (AFTN), referencing AIP SUP 014/24 and stating:

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(A1482/24 NOTAMN
Q) LFXX/QMRLC/IV/NBO/A/000/999/4337N00122E005
A) LFBO B) 2411040600 C) 2412191800
E) RWY 26 THR DISPLACED BY 320M DUE TO DRAINAGE REHABILITATION WORKS.
DECLARED DISTANCES: TORA 2680M, TODA 2830M, ASDA 2680M, LDA 2680M.
TEMPORARY RESA 120M ESTABLISHED. PAPI 26 REALIGNED TO 3.0 DEG.
MANDATORY COMPENSATORY MEASURES APPLY: DRY RWY ONLY, MAX XWIND 15KT.
REF AIP SUP 014/24.)
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C. Mandatory Airline Flight Operations Briefing

Prior to flight release, airline dispatchers must ensure flight crews receive the dedicated *Runway 26 Works Safety Briefing Package*, verifying that aircraft take-off and landing performance calculations account for the reduced LDA and dry-runway dispatch requirement.

9. ENFORCEMENT, AUDIT SCHEDULE & EXPIRY CONDITIONS

This Operational Exemption is valid strictly from **04 November 2024 at 06:00 UTC** until **19 December 2024 at 18:00 UTC** (a duration of 45 calendar days), unless formally revoked or suspended by the National Competent Authority.

- **Daily Joint Inspections:** Joint airside inspections between the aerodrome operations supervisor, ATC watch manager, and resident civil engineer must be executed and recorded at 05:00 UTC and 17:00 UTC daily.
- **Zero-Tolerance Revocation:** In the event of any runway excursion, pavement surface breach, or failure of worksite illumination, the competent authority shall immediately suspend operations under this exemption.

EASA AERODROMES REGULATORY DIRECTIVE
Validation Ref: EASA.ADR.2024-EXEMP-009-S3
Statutory Framework: Regulation (EU) 2018/1139 Article 71(1)

COMPETENT AUTHORITY SIGN-OFF
Directorate of Air Traffic Management & Aerodromes
APPROVED & SIGNED • 04 NOV 2024

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